

Geotechnical Investigation Brief

Project: Fully underground civil shelter, private plot, Rusanivski Sady, Kyiv Purpose of this brief: to be issued to a licensed geotechnical investigation contractor as a scope-of-work request. It defines what to find out, not the answers — no soil parameters are assumed or estimated here.

Note: this brief should be reviewed and, if needed, amended by the licensed structural engineer or a geotechnical professional before being sent to a contractor — particularly the number, depth, and location of test points, which should reflect their judgment and any local geological records they hold for the Rusanivski Sady area.

1. Site

- Location: Kyiv, Rusanivski Sady, private plot (exact cadastral boundary and address to be supplied by client).
- Proposed structure: fully underground, single-story reinforced-concrete box, approximate footprint on the order of a few tens of square meters, founded at an approximate depth of ~3 m below existing ground level (to be confirmed once a topographic survey — Doc.05 — establishes actual ground level).
- Known constraints: private residential plot; proximity to existing structures, trees, and underground utilities is not yet documented (see Doc.05).

2. Objective

Provide the data needed for a licensed structural engineer to:

- 1 Design the foundation slab and check bearing capacity.
- 2 Check the structure against uplift/flotation (buoyancy) under groundwater loading.
- 3 Determine lateral earth pressure on the below-grade walls.
- 4 Assess groundwater control / dewatering needs during construction.
- 5 Assess soil aggressivity toward concrete and steel reinforcement.

3. Information requested (deliverables from the geotechnical contractor)

#	Item	Why it's needed
1	Soil stratigraphy / lithology to a depth sufficient to cover the founding level plus an adequate margin (contractor to propose, typically several meters below the anticipated slab underside)	Bearing design, excavation planning
2	Groundwater level — including seasonal high-water estimate if available	Uplift/UPL check, waterproofing strategy, dewatering
3	Unit weight (bulk and saturated) of relevant soil layers	Lateral earth pressure, uplift calculation
4	Angle of internal friction (and cohesion, if relevant) of relevant soil layers	Lateral earth pressure calculation

#	Item	Why it's needed
5	Allowable/design bearing resistance (R0 or equivalent per applicable code) at the anticipated founding depth	Foundation slab design
6	Soil and groundwater aggressivity toward concrete and steel (sulfates, chlorides, pH, etc.)	Concrete durability class, reinforcement cover, waterproofing detailing
7	Recommendations on excavation support, slope stability, and dewatering method for the anticipated depth	Construction method statement, cost estimating
8	Any evidence of made ground, organic soils, or other conditions requiring special foundation treatment	Risk flag for the structural engineer

4. Suggested scope of investigation (for contractor to confirm/revise)

- Number and location of boreholes or test pits: to be proposed by the geotechnical contractor based on plot size and local geology; at minimum enough points to characterize variability across the footprint.
- In-situ testing (e.g., SPT, CPT, or equivalent) as appropriate to the contractor's standard practice.
- Laboratory testing of recovered samples for classification, strength, and chemical aggressivity.
- Groundwater monitoring: at least one reading at time of drilling; a standpipe/piezometer installation is recommended if seasonal variation is a concern, given the "worst case = GWL at ground level" scenario flagged as a risk in the design basis register.

5. Deliverable format requested

- A signed/sealed geotechnical investigation report suitable for use by a licensed structural engineer, including borehole logs, laboratory results, groundwater observations, and engineering recommendations (bearing resistance, lateral pressure parameters, aggressivity class).

6. Explicitly not requested from this brief

- No structural design, slab thickness, or reinforcement recommendation — that is the structural engineer's responsibility, using this report as an input.